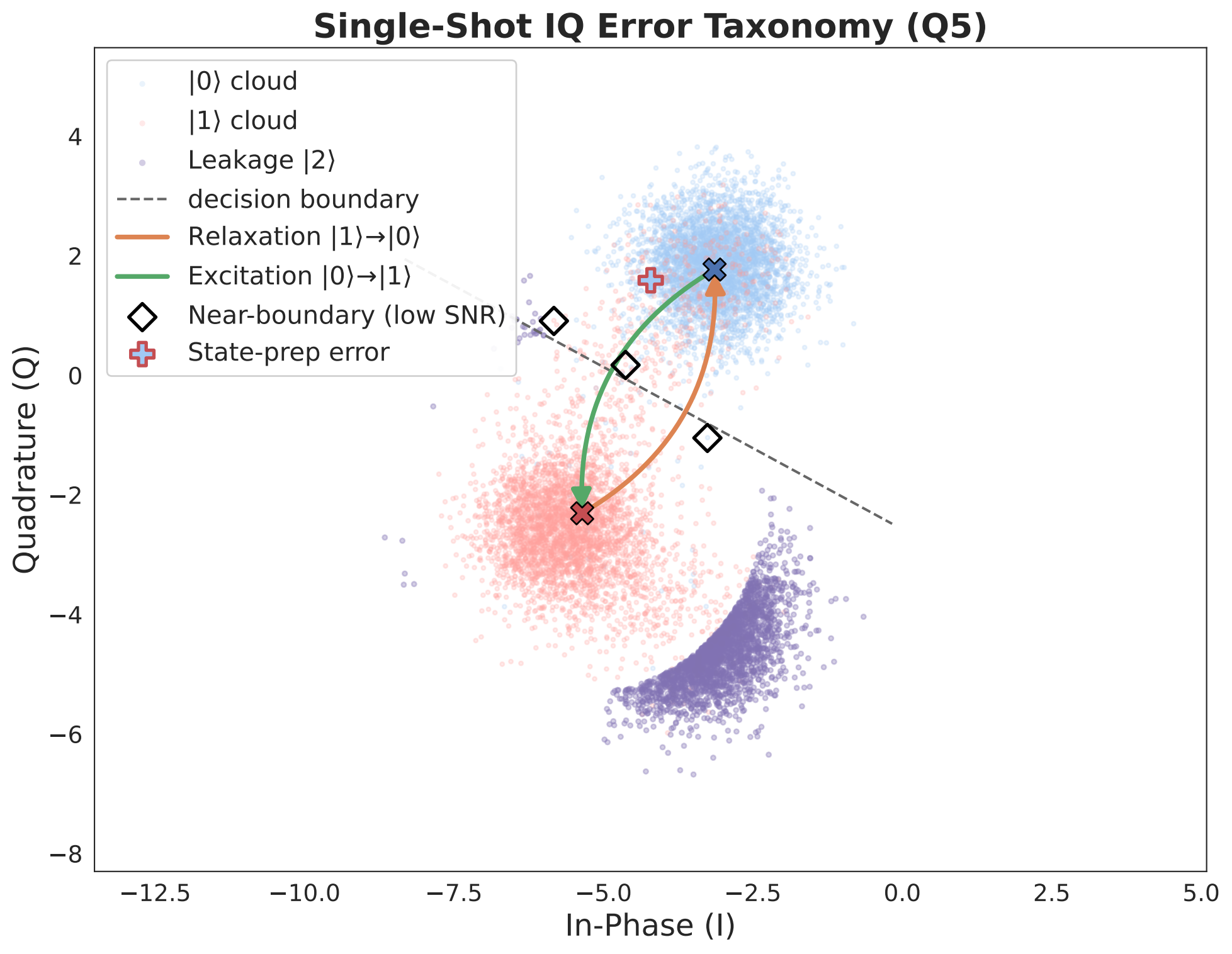

Single-Shot IQ Error Taxonomy (Q5)
|0⟩ cloud
|1⟩ cloud
Leakage |2⟩
decision boundary
Relaxation |1⟩→|0⟩
Excitation |0⟩→|1⟩
Near-boundary (low SNR)
State-prep error
Quadrature (Q)
In-Phase (I)
4
2
0
−2
−4
−6
−8
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0